

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

ReUseHub – Reuse and Redistribution System

Requirements Baseline v1.0, updated to v1.1 via Change Request CR-001 (Notification System)

Item	Details
Project	ReUseHub – Reuse and Redistribution System
Document	Software Requirements Specification (SRS)
Version	1.1 (baseline v1.0 + CR-001)
Baseline	Requirements Baseline v1.0, revised under CR-001 to v1.1
Development Approach	Incremental Development with Controlled Change Management
Technology	HTML, CSS, JavaScript, PHP, MySQL
Version Control	Git and GitHub
Status	Proposed for Review
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Date	29/08/2026

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the requirements for Version 1.0 of ReUseHub, a web-based reuse and redistribution system for university students and staff.

The document establishes the agreed requirements that will be used as the reference for system analysis, system design, database design, software development, testing, release activities, and Software Change Management.

The requirements contained in this document represent the initial proposed scope of the system. Once reviewed and approved, this document became the Requirements Baseline v1.0. A subsequent Change Request, CR-001, was raised to introduce a Notification System, and the resulting revised requirements are recorded in Section 23 of this document as Requirements Baseline v1.1.

1.2 Project Background

Students and staff often have usable items — books, stationery, electronics, or furniture — that they no longer need. These items are frequently kept unused or thrown away, even though other members of the university community may need them. There is currently no simple platform within the university to connect people who have unwanted items with people who are looking for them, resulting in unnecessary waste and missed opportunities for reuse.

ReUseHub is intended to provide a centralized, simple, web-based platform where students and staff can post unwanted items and others can browse, search, and request them.

2. Problem Statement

- Usable items owned by students and staff are frequently discarded instead of reused.
- There is no centralized university platform to connect people who have unwanted items with people who need them.
- Locating second-hand items within the university currently relies on informal word-of-mouth methods.
- There is no simple way to track whether an item is still available, already requested, or already given away.
- Users are not notified when something relevant to them happens on the platform, so they must repeatedly check the site manually.

3. Project Objectives

- 1. To develop a simple system for reusing unwanted items within the university.
- 2. To allow users to post items they no longer need.
- 3. To help users find useful second-hand items within the university community.
- 4. To reduce unnecessary disposal of usable items.
- 5. To provide a simple way for users to request available items.
- 6. To encourage sustainable practices by reducing waste through item reuse.
- 7. To provide an administrator with oversight of users, items, and requests through an admin dashboard.
- 8. To keep users informed of relevant activity through an in-system notification feature (introduced under CR-001).

4. Project Scope

4.1 In Scope for Version 1.0

Authentication

- User registration
- User login
- User logout
- Credential validation
- Secure password storage

Item Management

- Add item
- Edit / manage own items
- View and search available items
- View item details
- Update item status (Available, Requested, Given Away)

Requests

- Send item request
- Accept or reject requests
- View posted items and received / sent requests

Administration

- Admin dashboard to view all registered users
- Admin dashboard to view all posted items and their status
- Admin ability to view / moderate requests

Technical Scope

- HTML, CSS, JavaScript frontend
- PHP backend
- MySQL database
- Git and GitHub version control

4.2 Out of Scope for Version 1.0

- Payment or paid transactions (ReUseHub is a free reuse/donation platform, not a marketplace — items are exchanged in person)
- Mobile application (native)
- Advanced role-based access control
- Notification system (raised separately and approved under CR-001 — see Section 23)
- Production-scale deployment

5. Stakeholders

Stakeholder	Role	Interest / Responsibility
Student / Staff User	End User	Post items, browse items, send and manage requests
Administrator	System User	Manage users, items, and requests; monitor system activity
University / Institute	Management	Monitor adoption and encourage sustainable reuse practices
Project Team	Development Team	Analyze, design, develop, test, and maintain the system
Lecturer / Supervisor	Academic Supervisor	Monitor project progress and approve Change Requests

6. User Roles

6.1 Administrator

The administrator manages the overall system through an admin dashboard. The administrator shall be able to:

- Log in and log out
- View the list of all registered users
- View all posted items and their current status
- View all requests raised in the system
- Update or remove inappropriate items if required
- Mark an item as Given Away on behalf of a user, where necessary

6.2 Registered User (Student / Staff)

A registered user posts and requests items. The user shall be able to:

- Register and log in / log out
- Add, edit, and manage their own items
- Search and view available items
- Send a request for an item
- Accept or reject requests received on their own items
- Update the status of their own item (Available, Requested, Given Away)
- View their posted items and their sent / received requests

7. Functional Requirements (Version 1.0)

Functional requirements describe what the system shall do.

7.1 Authentication Requirements

FR-01 – User Registration

The system shall allow a new user to register using a username, email, and password.

FR-02 – User Login

The system shall allow a registered user to log in using valid credentials.

FR-03 – Login Validation

The system shall validate the credentials provided by the user and display an appropriate error message if invalid.

FR-04 – User Logout

The system shall allow an authenticated user to log out of the system.

7.2 Item Management Requirements

FR-05 – Add Item

The system shall allow an authenticated user to post a new item with a name, description, category, image, and the seller's contact number, so that an interested user can arrange to collect the item in person.

FR-06 – Manage Item

The system shall allow a user to edit or remove their own posted items.

FR-07 – View / Search Items

The system shall allow users to view and search available items by name or category.

FR-08 – View Item Details

The system shall allow a user to view the full details of a selected item, including the seller's contact number while the item is Available.

FR-09 – Update Item Status

The system shall allow the item owner (or an administrator) to update an item's status to Available, Requested, or Given Away.

FR-09A – Hide Contact Number on Given Away

The system shall hide the seller's contact number from the item details once the item's status is updated to Given Away, so that the number is no longer shown to other users.

7.3 Request Management Requirements

FR-10 – Send Item Request

The system shall allow a user to send a request for an item that is marked Available.

FR-11 – Accept / Reject Request

The system shall allow an item owner to accept or reject a received request.

FR-12 – View Requests

The system shall allow a user to view their posted items along with requests they have sent and received.

7.4 Administration Requirements

FR-13 – Admin Dashboard

The system shall provide an administrator with a dashboard showing all registered users, all posted items with their status, and all requests.

FR-14 – Admin Item Oversight

The system shall allow an administrator to update or remove any item and, where necessary, mark it as Given Away.

8. Non-Functional Requirements

8.1 Security

NFR-01 – Password Security

The system shall store passwords using secure password hashing rather than plain text.

NFR-02 – Authentication

Protected system functions shall require the user to be authenticated.

8.2 Usability

NFR-03 – User Interface

The system shall provide a simple, understandable, and consistent user interface.

NFR-04 – Validation Messages

The system shall provide clear messages when input is invalid or an operation cannot be completed.

8.3 Reliability & Performance

NFR-05 – Response Time

Commonly used pages should respond within approximately three seconds under normal conditions.

NFR-06 – Data Integrity

The system shall maintain the integrity and consistency of stored item and request data.

8.4 Portability & Responsiveness

NFR-07 – Browser Compatibility

The web application should operate correctly on commonly used modern web browsers.

NFR-08 – Responsive Interface

The user interface should adapt to common desktop, tablet, and mobile screen sizes.

9. User Stories

ID	Title	Story
US-01	Registration	As a new user, I want to register an account, so that I can access the system.
US-02	Login	As a registered user, I want to log in, so that I can use the functions available to me.
US-03	Add Item	As a user, I want to post an item I no longer need, so that others can find and request it.
US-04	Search Item	As a user, I want to search for items, so that I can quickly find something I need.
US-05	Send Request	As a user, I want to request an available item, so that I can obtain it.
US-06	Accept / Reject Request	As an item owner, I want to accept or reject requests, so that I can decide who receives my item.
US-07	Update Item Status	As an item owner, I want to update my item's status, so that other users know whether it is still available.
US-08	Admin Dashboard	As an administrator, I want to view all users, items, and requests, so that I can monitor and manage the system.
US-08A	Contact Seller	As a user, I want to see the seller's contact number on an available item, so that I can arrange to meet and collect it in person.

10. Acceptance Criteria

AC-01 – Registration

Given a new visitor is on the registration page, when valid details are submitted, then an account shall be created and a success message displayed.

AC-02 – Login

Given a registered user is on the login page, when valid credentials are entered, then the user shall be authenticated and redirected to the dashboard. Invalid credentials shall show an error.

AC-03 – Add Item

Given an authenticated user, when valid item details are submitted, then the item shall be saved and appear in the item list with status Available.

AC-04 – Send Request

Given an item is Available, when a user sends a request, then the request shall be recorded and the item owner notified in their requests list.

AC-05 – Accept / Reject Request

Given a request exists on the owner's item, when the owner accepts or rejects it, then the request status shall update accordingly and, on acceptance, the item status shall update to Requested or Given Away as configured.

AC-06 – Admin Dashboard

Given an administrator is logged in, when the dashboard is opened, then all users, items, and requests shall be displayed.

AC-06A – Contact Number Visibility

Given an item is Available, when a user views its details, then the seller's contact number shall be displayed. Given the item is later marked Given Away, when the item details are viewed again, then the contact number shall no longer be shown.

11. MoSCoW Prioritization

Priority	Requirements
Must Have	Registration, Login, Logout, Add/Edit Item, Item Status Update, Send Request, Accept/Reject Request, Admin Dashboard
Should Have	Item Search, Improved Validation, User Feedback Messages
Could Have	Item Categories / Filters
Won't Have in v1.0	Notification System (introduced later via CR-001)

12. Product Backlog

ID	Item	Priority	Related Req.	Status
US-01	Registration	Must	FR-01	Planned
US-02	Login / Logout	Must	FR-02, FR-03, FR-04	Planned
US-03	Add / Manage Item	Must	FR-05, FR-06	Planned
US-04	Search / View Items	Must	FR-07, FR-08	Planned
US-05	Send Request	Must	FR-10	Planned
US-06	Accept / Reject Request	Must	FR-11	Planned

ID	Item	Priority	Related Req.	Status
US-07	Update Item Status	Must	FR-09	Planned
US-08	Admin Dashboard	Must	FR-13, FR-14	Planned
—	Notification System	Won't Have (v1.0) → Approved (v1.1)	See CR-001	Future / In Progress

13. Use Cases

Use Case ID	Use Case	Primary Actor
UC-01	Register / Login	User
UC-02	Add Item	User
UC-03	Search / View Items	User
UC-04	Send Request	User
UC-05	Accept / Reject Request	User
UC-06	Update Item Status	User / Admin
UC-07	View Admin Dashboard	Administrator

14. Business Rules

- BR-01 – Only authenticated users may post, request, or manage items.
- BR-02 – A request may only be sent for an item that is marked Available.
- BR-03 – Only the item owner or an administrator may change an item's status.
- BR-04 – Passwords must not be stored in plain text.
- BR-05 – Each user account shall have a unique email address.
- BR-05A – The seller's contact number shall only be visible on an item while its status is Available; once the item is marked Given Away, the contact number shall be hidden from other users.

15. Data Requirements

15.1 User Information

- User ID
- Name
- Email
- Password (hashed)
- Role (User / Admin)

15.2 Item Information

- Item ID
- Owner (User ID)
- Item Name
- Description
- Category
- Image
- Seller Contact Number (hidden once Given Away)
- Status (Available / Requested / Given Away)

15.3 Request Information

- Request ID
- Item ID
- Requester (User ID)
- Request Status (Pending / Accepted / Rejected)
- Request Date

16. System Constraints

- 1. The application will be developed as a web-based system.
- 2. HTML, CSS, and JavaScript will be used for the frontend.
- 3. PHP will be used for the backend.
- 4. MySQL will be used as the database.
- 5. Git and GitHub will be used for version control.
- 6. Version 1.0 focuses on core posting, browsing, and requesting functionality.
- 7. New requirements after baseline approval must follow the Change Request process.

17. Assumptions

- Users will have access to a modern web browser.
- Users will register with a valid university email address.
- The development environment will use XAMPP (Apache, PHP, MySQL).
- Requirements may evolve after Version 1.0 and will be managed through Software Change Management.

18. Requirements Traceability Matrix

Requirement	User Story	Acceptance Criteria
FR-01	US-01	AC-01
FR-02/03/04	US-02	AC-02
FR-05/06	US-03	AC-03
FR-10	US-05	AC-04
FR-11	US-06	AC-05
FR-13/14	US-08	AC-06

19. Requirements Review

Q: Is a notification feature included in Version 1.0?

Decision: No. It was not part of the original proposal and is introduced later through CR-001 (see Section 23).

Q: Is payment / selling of items included?

Decision: No. ReUseHub is a free reuse and redistribution platform, not a marketplace.

Q: Can a regular user access the admin dashboard?

Decision: No. The admin dashboard is restricted to the administrator role.

20. Version 1.0 Requirements Baseline

After stakeholder and supervisor review, the requirements contained in Sections 1–19 of this document were approved and established as Requirements Baseline v1.0 for ReUseHub.

After baselining, any new requirement is submitted through the Software Change Management process using a formal Change Request (CR), rather than being added directly to the approved scope.

21. Configuration and Version Control

Item	Value
Baseline Name	Requirements Baseline v1.0
Git Tag	requirements-v1.0
Status	Approved after Sign-off
Baseline Date	_____

22. Change Management Process

Once Version 1.0 is baselined, new requirements are handled as follows:

Customer / Supervisor Request → Change Request (CR) → Impact Analysis → Change Approval → Requirements Update → Design Update → Database Update → Backend Update → Frontend Update → Testing → Documentation Update → New Version / Release

23. Change Request CR-001 – Notification System

Following approval of Requirements Baseline v1.0, the supervisor requested that an additional feature be introduced: an in-system Notification feature, so that users are informed of relevant activity (such as new items and updates to their requests) through a notification icon, instead of having to check the site manually. This section documents CR-001 in full and records the resulting Requirements Baseline v1.1.

23.1 Change Request Form

Field	Details
CR ID	CR-001
Title	Introduce a Notification System
Raised By	Supervisor, during project presentation review
Description	Add a notification icon to the interface that alerts a user to relevant activity — new items posted, status changes on requested items, and updates on requests they have sent or received.
Reason	Users currently have no way of knowing about relevant activity without manually revisiting the site; a notification feature improves usability and engagement.
Affected Baseline	Requirements Baseline v1.0
Priority	Should Have
Status	Approved

23.2 Impact Analysis

Area	Impact
Requirements	New functional requirements FR-15 to FR-19 added (Section 23.3).
Database Design	New notifications table required (recipient, message, type, related item/request, read status, timestamp).
Backend	New logic to create a notification whenever a new item is posted, a request is sent, a request is accepted/rejected, or an item's status changes.
Frontend	New notification icon with unread-count badge and a dropdown / page listing notifications.
Testing	New test cases required to confirm notifications are created and displayed correctly for each trigger event.

Area	Impact
Schedule / Effort	Low to moderate — feature is additive and does not change existing item or request workflows.

23.3 Updated Functional Requirements (Version 1.1)

FR-15 – Notification Icon

The system shall display a notification icon in the interface showing the count of unread notifications for the logged-in user.

FR-16 – Notify on New Request

The system shall create a notification for an item owner when another user sends a request for their item.

FR-17 – Notify on Request Decision

The system shall create a notification for a requester when their request is accepted or rejected.

FR-18 – Notify on Item Status Change

The system shall create a notification for a requester when an item they requested changes status to Given Away.

FR-19 – Notify on New Item Posted

The system shall create a notification for registered users when a new item is posted, so that users can discover newly available items.

FR-20 – View / Mark Notifications

The system shall allow a user to view their notifications and mark them as read.

23.4 Updated User Stories & Acceptance Criteria

ID	Story
US-09	As a user, I want to see a notification when someone requests my item, so that I can respond promptly.
US-10	As a user, I want to be notified when my request is accepted or rejected, so that I know the outcome.
US-11	As a user, I want to see a badge on the notification icon, so that I know I have unread updates.

AC-07 – Notification on Request

Given a user sends a request for an item, when the request is created, then the item owner shall receive a notification and the unread count on their notification icon shall increase by one.

AC-08 – Notification on Decision

Given an item owner accepts or rejects a request, when the decision is saved, then the requester shall receive a notification reflecting the outcome.

AC-09 – Mark as Read

Given a user opens their notification list, when they view or select a notification, then it shall be marked as read and the unread count shall decrease accordingly.

23.5 Updated Business Rules

- BR-06 – A notification shall be linked to the user it is intended for and shall not be visible to other users.
- BR-07 – A notification shall be marked as read once the user has viewed it.

23.6 Updated Data Requirements — Notification Information

- Notification ID
- Recipient (User ID)
- Type (New Item / Request Received / Request Decision / Item Given Away)
- Related Item / Request ID
- Message Text
- Read Status
- Created Date/Time

23.7 Updated MoSCoW Prioritization

Priority	Requirements
Must Have (v1.1)	All Version 1.0 Must Have items
Should Have (v1.1)	Notification System (FR-15 to FR-20) — approved under CR-001
Could Have	Item Categories / Filters

24. Version 1.1 Requirements Baseline (Post-CR-001)

Following impact analysis and approval of CR-001, the requirements in Section 23.3 (FR-15 to FR-20) are incorporated into the system scope. The combined Version 1.0 requirements together with the CR-001 additions constitute Requirements Baseline v1.1 for ReUseHub.

Item	Value
Baseline Name	Requirements Baseline v1.1
Change Request	CR-001 – Notification System
Git Tag	requirements-v1.1
Status	Approved after Sign-off
Baseline Date	_____

25. Version History

Version	Date	Description	Status
0.1	_____	Initial requirements draft	Draft
1.0	_____	Requirements baseline established	Approved
1.1	_____	CR-001 (Notification System) analyzed and merged into baseline	Approved

26. Approval

By approving this document, the relevant stakeholders confirm that the requirements defined, including the Version 1.1 update introduced under CR-001, represent the agreed scope of ReUseHub.

Role	Name	Signature	Date
Student / Project Team	_____	_____	_____
Lecturer / Supervisor	_____	_____	_____